ELM: 4 mm pitch, 2.5 mm holes
Xe–He pilot

NAUSICAA

Setup

Xe–15% He only: the same 100 bb0nu source tracks and primary-pair realizations as the preceding pilot. Electron reference drift is sampled uniformly over 10–150 cm using the same deterministic placement stream; the saved reference and current events have identical IDs and drift distances. Ion distance is its complement within 150 cm. Original hit coordinates are retained, and each event uses its reference distance for diffusion and attachment. Only electron photon collection and response are replayed.

The square hole pitch is 4 mm; hole and guide diameter are 2.5 mm. EL depth is 5 mm at 15 bar and E/P = 2 kV/cm/bar, with unit electron collection. Guides have a 0.5 mm radial xenon gap. Uniform axial EL and absorbing walls give a recomputed exit acceptance. TPB conversion 0.65, forward factor 0.5 and guide trapping $1-1/1.5$ apply once, giving emitted-VUV to saved-blue-photon probability 0.01187461. Propagation and NOA 87 coupling are ideal.

S14160-3015PS remains 3×3 mm^2 with 39,984 physical cells, PDE 0.32, gain 360,000, +4 V and 25 C. Uniform illumination of the 2.5 mm circular guide footprint uses 21,808 cells, rounded from $39984 \times \pi \times 1.25^2 / 9$. The remaining 18,176 cells receive dark triggers. The full-device 700 kcps dark rate is split by cell count. The validated statistical recovery model runs at 128 substeps per 250 ns bucket, with 12.5 and 25 ns provisional recovery constants. These are sensitivity inputs rather than measured recharge constants. PDE is the packaged peak-value approximation; crosstalk, afterpulsing and gain spread remain deferred.

Losses and retained fraction

Pitch / hole diameter	Recovery	SiPM loss (%)	ASIC loss (%)	Events clipped (%)	Retained / 100
5 mm / 3 mm hole	12.5 ns	0.0790	0.2184	11.0	89
4 mm / 2.5 mm hole	12.5 ns	0.0673	0.0165	3.0	97
5 mm / 3 mm hole	25 ns	0.1576	0.2098	10.0	90
4 mm / 2.5 mm hole	25 ns	0.1343	0.0136	3.0	97

Corrected event widths

Pitch / hole diameter	Recovery	Detected RMS (%)	SiPM RMS (%)	Unclipped RMS (%)	ASIC RMS (%)	Retained ASIC RMS (%)
5 mm / 3 mm hole	12.5 ns	0.1807	0.1951	0.2347	0.9396	0.2258
4 mm / 2.5 mm hole	12.5 ns	0.1866	0.2024	0.2942	0.3360	0.2928
5 mm / 3 mm hole	25 ns	0.1683	0.2107	0.2564	0.9521	0.2222
4 mm / 2.5 mm hole	25 ns	0.1943	0.2376	0.2918	0.3349	0.2777

Pitch / hole diameter	Recovery	Event-mean avg PE	Event-max avg PE	Largest count	Largest charge PE
5 mm / 3 mm hole	12.5 ns	121.65	1,994.4	8,738	8,614.5
4 mm / 2.5 mm hole	12.5 ns	76.23	1,249.8	5,517	5,449.1
5 mm / 3 mm hole	25 ns	121.64	1,986.4	8,672	8,445.6
4 mm / 2.5 mm hole	25 ns	76.26	1,251.4	5,488	5,351.8

Definitions and event selection

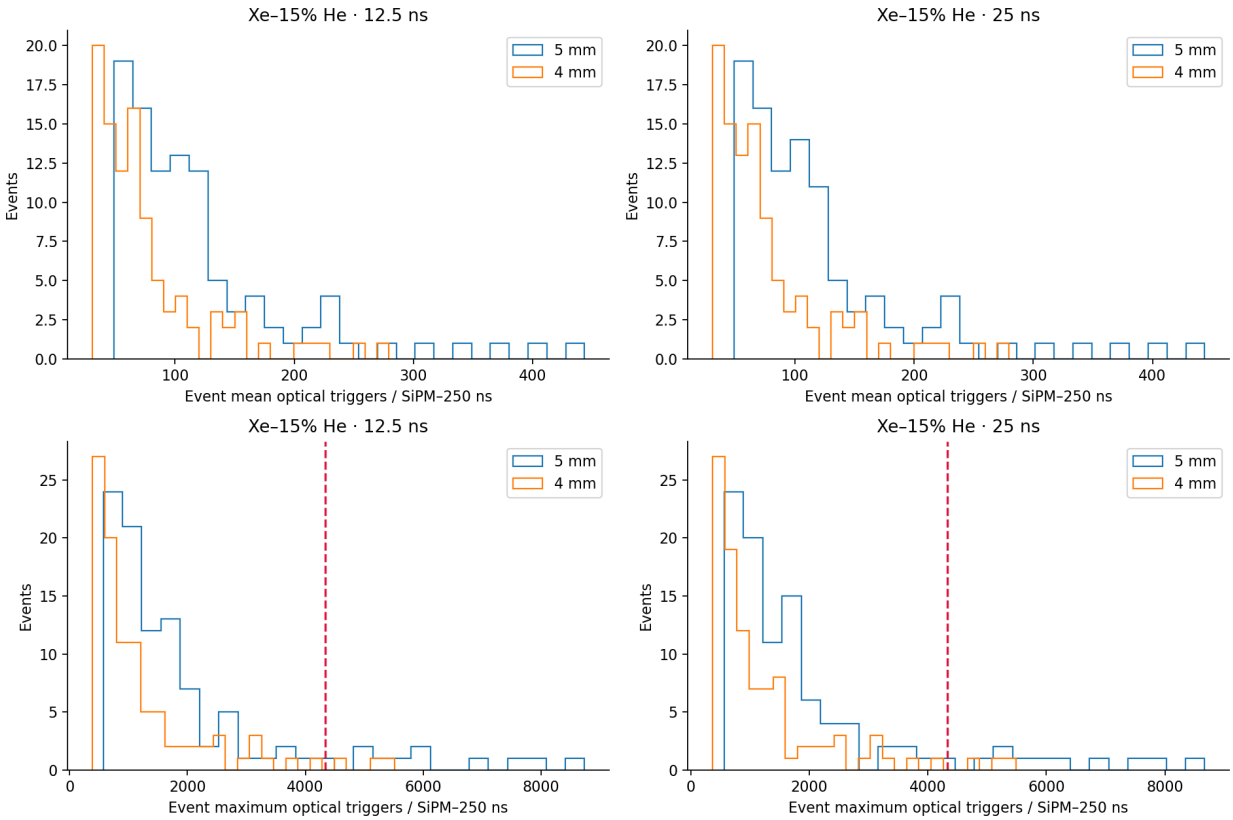
Optical trigger counts are sampled PDE successes; recurrent firing is allowed. Charge-equivalent PE is collected charge divided by e × gain. Distinct optical occupancy is the conditional expected number of cells reached by the triggers, divided by the illuminated count. Mean/max comparisons select the same photon-positive input cells at every stage. Electronics adds 3.5417 PE RMS Gaussian noise per bucket and clips at 1 mA (4334.38 charge-equivalent PE). Signed values are retained. SiPM loss compares optical charge with optical triggers; ASIC loss compares clipped charge with the incoming sensor charge. Event widths are full-distribution RMS/mean after known attachment and deposited-energy correction. Electronics totals subtract the expected dark baseline and include all acquired cells on illuminated channels plus four quiet buckets at either end.

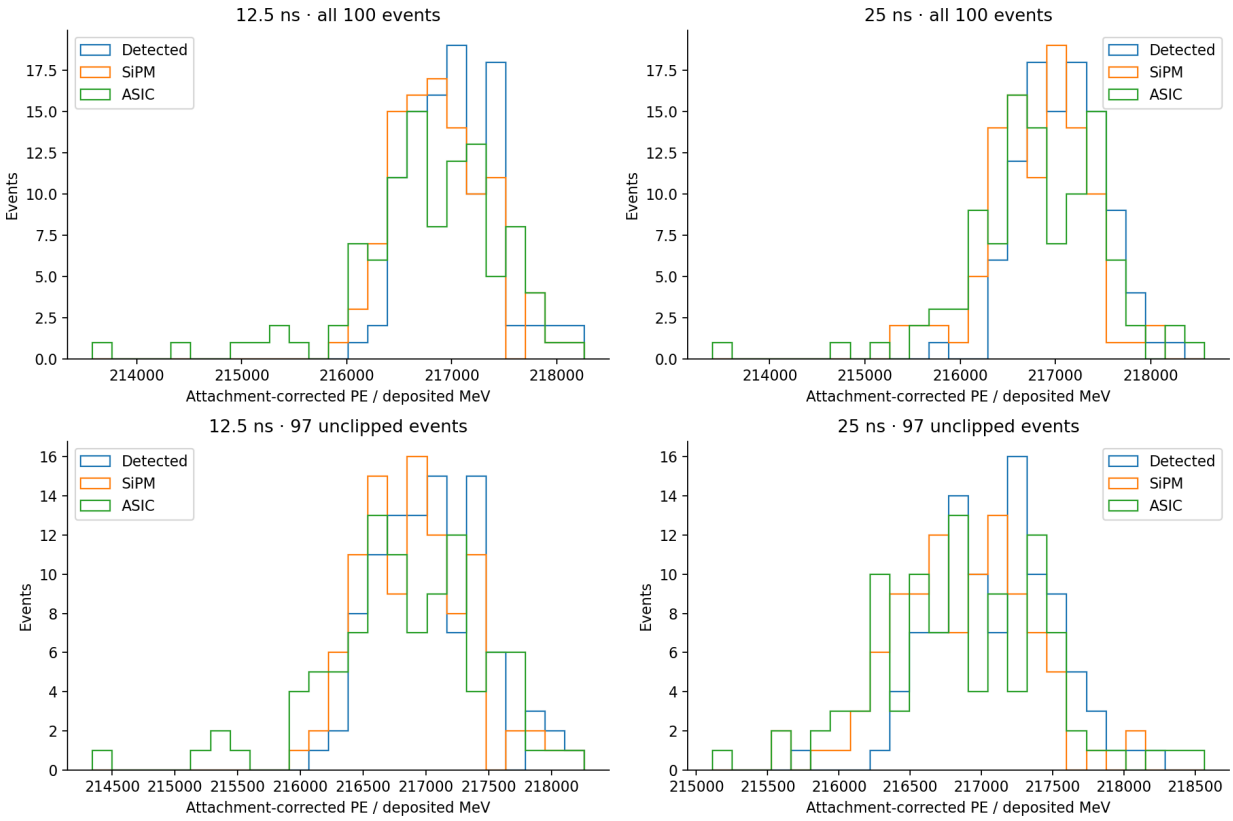
The retained sample rejects an entire event when any acquired bucket clips at the ASIC. The table reports both its efficiency and the resulting conditional energy width. This pilot uses the simulated clipping flag; it is not an energy correction. No amplitude or signal-to-noise cut is applied. Both recovery cases reuse the same blue photons with independent response draws, and the 5 mm comparison uses saved results without new simulation.

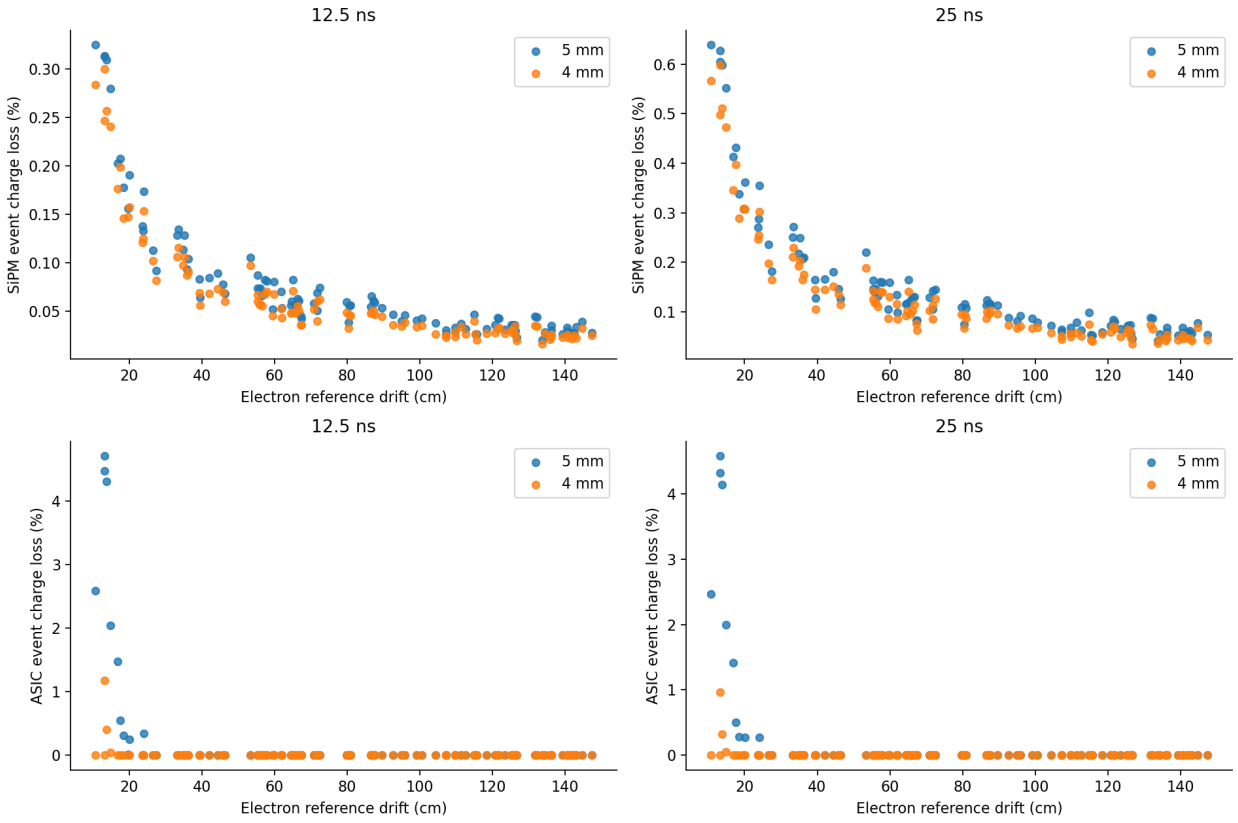
Timing and validation

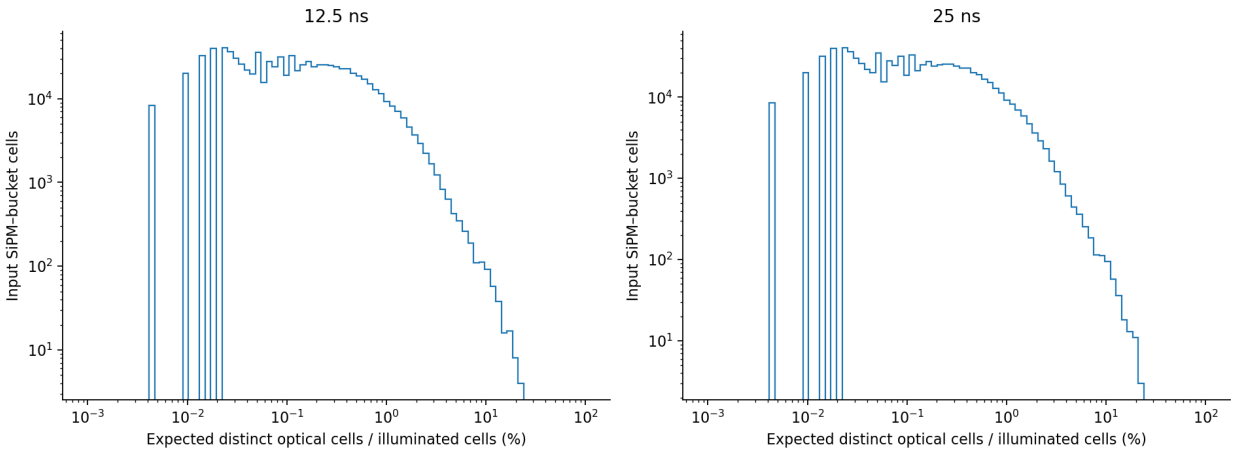
Photon replay/archive: 4.20 s; 12.5 ns response: 19.56 s; 25 ns response: 16.75 s; combined: 40.51 s. Timings include stage compilation and I/O; process startup and presentation are separate.

Thirty-one tests cover selected-gas cards, geometry, placement, illuminated cell count, dark-rate conservation, charge bounds, recurrent firing and substep convergence. Saved-event checks verify primary checksums, exact matching of IDs and placements against 5 mm, PDE sampling, charge ledgers and quiet noise. Rare second-moment feasibility repairs preserve sampled charge and the first moment; their maximum size and frequency are bounded and recorded. No charge-variance limits occurred.


Xe–15% He · 12.5 ns
Xe–15% He · 25 ns
5 mm
4 mm
Events
Event mean optical triggers / SiPM–250 ns
Event maximum optical triggers / SiPM–250 ns


12.5 ns · all 100 events
25 ns · all 100 events
12.5 ns · 97 unclipped events
25 ns · 97 unclipped events
Detected
SiPM
ASIC
Events
Attachment-corrected PE / deposited MeV


12.5 ns
25 ns
5 mm
4 mm
SiPM event charge loss (%)
ASIC event charge loss (%)
Electron reference drift (cm)


12.5 ns
25 ns
Input SiPM-bucket cells
Expected distinct optical cells / illuminated cells (%)

Largest event losses

Recovery	Event ID	Drift (cm)	Maximum triggers	Occupancy (%)	SiPM loss (%)	ASIC loss (%)
12.5 ns	78	13.21	5,517	22.35	0.300	1.178
12.5 ns	67	13.75	5,220	21.29	0.257	0.403
12.5 ns	2	10.78	4,195	17.50	0.284	0.000
12.5 ns	54	14.87	4,568	18.90	0.240	0.043
12.5 ns	52	13.17	3,826	16.09	0.246	0.000
25 ns	78	13.21	5,488	22.25	0.598	0.967
25 ns	67	13.75	5,179	21.14	0.511	0.325
25 ns	2	10.78	4,235	17.65	0.567	0.000
25 ns	54	14.87	4,717	19.45	0.473	0.052
25 ns	52	13.17	3,819	16.06	0.498	0.000

Validation

Recovery	Ledger checks	State repairs	Variance limits	Max repair ppm	PDE z	Quiet RMS PE
12.5 ns	100	1	0	0.000	-0.98	3.5420
25 ns	100	1	0	0.000	1.56	3.5421

Reproduction

`results/elm_sipm_pitch4mm_pilot/elm_sipm_pitch4mm_pilot.h5`; file index `results/elm_sipm_pitch4mm_pilot/files.toml`. Run `make elm-pitch4-pilot`; presentation only `make elm-pitch4-report`.

Input cards

```
schema = "nausicaa.electron_detector_spec.v1"
pressure_bar = 15.0
helium_fraction = 0.15
drift_speed_mm_us = 1.0
lifetime_ms = 15.0
hole_pitch_mm = 4.0
hole_diameter_mm = 2.5
hole_depth_mm = 5.0
sipm_shape = "square"
sipm_side_mm = 3.0
bucket_width_us = 0.25
reduced_field_kV_cm_bar = 2.0
j_variance_ratio = 0.030
j_status = "user_specified"
j_reference = "Oliveira Figure 6; corrected approximate user reading, 2026-11-04"

optical_model = "elm_blue_meridional"
electron_collection_efficiency = 1.0

[optics]
photon_band = "blue_tpb"
tpb_conversion_efficiency = 0.65
forward_fraction = 0.5
guide_refractive_index = 1.5
gap_refractive_index = 1.0
guide_length_mm = 100.0
gas_gap_radial_mm = 0.5
propagation_transmission = 1.0
```

```
coupling_transmission = 1.0
electrode_transmission = 1.0
coupling_adhesive = "Norland NOA 87"
guide_length_policy = "fixed_grading_field"
grading_field_kV_cm = 4.65
cathode_potential_V = 1200.0
entry_potential_V = 31500.0
exit_potential_V = 46500.0
sensor_plane_potential_V = 0.0
reference = "docs/elm.tex; isotropic axial EL, absorbing walls, meridional isotropic-in-core
↪  capture"

[pure_xe]
d_T_mm_sqrtbar_per_sqrtcm = 3.5
d_L_mm_sqrtbar_per_sqrtcm = 0.9
yield_gain_photons_per_electron_kV = 137.0
yield_threshold_kV_cm_bar = 0.73

[xe_he15]
d_T_mm_sqrtbar_per_sqrtcm = 1.6
d_L_mm_sqrtbar_per_sqrtcm = 0.75
yield_gain_photons_per_electron_kV = 139.0
yield_threshold_kV_cm_bar = 0.80

[placement]
mode = "uniform_electron_reference"
minimum_electron_drift_cm = 10.0
maximum_electron_drift_cm = 150.0
seed = 2026
reference = "Event reference drift; common diffusion and attachment distance for all hits, original
↪  hit coordinates retained"

schema = "nausicaa.sipm_spec.v1"
response_model = "statistical"
statistical_substeps = 128
sparse_trigger_limit = 16
microcell_count = 39984
illuminated_microcell_count = 21808
microcell_pitch_m = 15e-6
recovery_s = 1.25e-08
arrival_model = "uniform_in_bucket"
illumination = "uniform_disk_subset"
recovery_pde = "constant"
initial_state = "stationary_dark"
charge_assignment = "avalanche_bucket"
input_photon_band = "blue_tpb"
shape = "square"
side_m = 0.003
pde = 0.32
gain = 360000.0
dark_reference_rate_Hz = 700000.0
dark_reference_area_m2 = 9e-6
crosstalk = false
afterpulsing = false
gain_spread = false
seed = 2026

[acquisition]
channels = "illuminated_in_event"
additional_channels = []
quiet_before_buckets = 4
quiet_after_buckets = 4
empty_first_bucket = 0
empty_last_bucket = 7

[provenance]
```

```
device = "Hamamatsu S14160-3015PS, 3x3 mm2; nominal +4 V, 25 C"
pde = "0.32 peak PDE at 460 nm used as provisional TPB-spectrum average; packaged PDE includes
↪  entrance window"
gain = "3.6e5 nominal electrons per primary avalanche; charge per fully recovered avalanche"
dark_rate = "S14160-3015PS nominal primary dark rate 700 kcps at 25 C and +4 V"
window = "S14160-3015PS silicone resin, nominal n=1.57; unit external coupling used in pilot"
datasheet = "papers/SiPM/Hamamatsu_S14160_small_cells.pdf"

recovery = "Provisional 12p5 ns exponential recovery; sensitivity parameter, not a measured 3015PS
↪  recharge constant"
illumination = "Uniform centered 2.5 mm disk on 3x3 mm square: round(39984*pi*1.25^2/9)=21808 lit
↪  cells; remaining 18176 dark-only; full 700 kcps dark rate split by cell count"

schema = "nausicaa.sipm_spec.v1"
response_model = "statistical"
statistical_substeps = 128
sparse_trigger_limit = 16
microcell_count = 39984
illuminated_microcell_count = 21808
microcell_pitch_m = 15e-6
recovery_s = 2.5e-08
arrival_model = "uniform_in_bucket"
illumination = "uniform_disk_subset"
recovery_pde = "constant"
initial_state = "stationary_dark"
charge_assignment = "avalanche_bucket"
input_photon_band = "blue_tpb"
shape = "square"
side_m = 0.003
pde = 0.32
gain = 360000.0
dark_reference_rate_Hz = 700000.0
dark_reference_area_m2 = 9e-6
crosstalk = false
afterpulsing = false
gain_spread = false
seed = 2026

[acquisition]
channels = "illuminated_in_event"
additional_channels = []
quiet_before_buckets = 4
quiet_after_buckets = 4
empty_first_bucket = 0
empty_last_bucket = 7

[provenance]
device = "Hamamatsu S14160-3015PS, 3x3 mm2; nominal +4 V, 25 C"
pde = "0.32 peak PDE at 460 nm used as provisional TPB-spectrum average; packaged PDE includes
↪  entrance window"
gain = "3.6e5 nominal electrons per primary avalanche; charge per fully recovered avalanche"
dark_rate = "S14160-3015PS nominal primary dark rate 700 kcps at 25 C and +4 V"
window = "S14160-3015PS silicone resin, nominal n=1.57; unit external coupling used in pilot"
datasheet = "papers/SiPM/Hamamatsu_S14160_small_cells.pdf"

recovery = "Provisional 25 ns exponential recovery; sensitivity parameter, not a measured 3015PS
↪  recharge constant"
illumination = "Uniform centered 2.5 mm disk on 3x3 mm square: round(39984*pi*1.25^2/9)=21808 lit
↪  cells; remaining 18176 dark-only; full 700 kcps dark rate split by cell count"

baseline_pe = 0.0
bucket_width_s = 2.5e-7
current_limit_A = 0.001
lower_clipping = false
noise = "independent_gaussian_pe"
noise_rms_pe = 3.5416666666666665
```

```
schema = "nausicaa.sipm_electronics_spec.v1"
seed = 2027
transfer = "bucket_average_current"

[provenance]
noise = "Same input charge RMS as 0.5 PE at gain 2.55e6; rescaled to gain 3.6e5"
range = "User design input: upper clipping at 1 mA; supersedes the provisional 300 PE limit"
timing = "User 0.25 us buckets; integrated charge divided by bucket duration"
```